DEVOPS TECHNICAL PROJECT REPORT

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CI/CD Pipeline with GitHub Actions & Docker (No Cloud Needed)

Automated Continuous Integration, Multi-Arch Containerization & Local Kubernetes Orchestration

AUTHOR	GITHUB REPOSITORY	DOCKER HUB IMAGE	LOCAL TARGET
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1 Abstract

Modern cloud computing services (such as AWS, Azure, and GCP) frequently incur recurring operational fees and complex credential barriers, making them inaccessible for rapid prototyping, local staging, academic verification, and cost-constrained deployments. This project architects and implements a production-grade, zero-cost Continuous Integration and Continuous Deployment (CI/CD) pipeline. By unifying GitHub Actions for cloud-hosted testing and automated container builds, Docker Hub as a free public artifact registry, and a local Minikube cluster as the deployment target, the system demonstrates an end-to-end delivery cycle without external cloud infrastructure costs.

> 💡 **Core Architecture Highlight**
>
> Developers commit code locally → GitHub Actions automatically triggers multi-version linting and unit tests → Docker Buildx builds a secure, non-root Alpine container image → Image is published to Docker Hub → Local Minikube pulls and deploys the container with zero-downtime rolling updates and active health probes.

2 Introduction

Continuous Integration and Continuous Deployment represents the foundation of modern DevOps methodologies, enforcing code consistency, automated verification, and rapid release cadences. However, organizations and developers often face steep friction when establishing deployment targets due to cloud vendor lock-in, infrastructure subscription requirements, and security token complexities.

The primary objective of this project is to eliminate cloud dependency while preserving strict industry best practices: automated quality gates, deterministic build caching, container immutability, container security via unprivileged runtime users, and Kubernetes-native liveness and readiness monitoring. The implementation features a modular Node.js microservice equipped with real-time diagnostic telemetry, container health probes, and an interactive web dashboard.

3 Tools & Technologies Used

TOOL / PLATFORM	CATEGORY	ROLE IN IMPLEMENTATION	KEY BENEFIT / VALUE
GitHub Actions	CI/CD Automation	Orchestrates automated workflows on git push events via Ubuntu runners.	Free hosted runner compute, secret protection & native test matrix.
Docker & Buildx	Containerization	Multi-stage Dockerfile packaging the app into a ~50MB Alpine image.	Immutable environments, non-root execution (UID 1000), layer caching.
Docker Hub	Container Registry	Hosts published container tags (latest, sha-tag) publicly for automated pull.	Free public repository auto-created on initial push.
Minikube & k8s	Local Orchestration	Runs a local single-node Kubernetes cluster hosting 2-replica deployments.	Enterprise-grade rolling updates, self-healing, NodePort routing.
Node.js (v20 / v22)	Application Runtime	Lightweight web service with built-in HTTP server, metrics & health probes.	Zero third-party runtime dependencies, ultra-fast sub-second boot time.

4 Steps Involved in Building the Project

Step 1: Microservice Development & In-Memory Test Harness

Application Layer

Engineered an event-driven web application (`src/app.js`, `src/server.js`) delivering an interactive UI, memory metrics, and health endpoints (`/health`, `/api/info`). Constructed an in-memory test suite (`tests/app.test.js`) using Node's native `node:test` and `node:assert` runner, executing all 7 unit tests in under 3 milliseconds without socket port conflicts.

Step 2: Optimized Containerization & Docker Compose Setup

Docker Engine

Authored an Alpine-based `Dockerfile` implementing defense-in-depth security: non-root user execution (`USER node`), layer caching for package manifests, and an internal `HEALTHCHECK` instruction. Configured `docker-compose.yml` mapping port `3000:3000` with health monitoring and restart policies for local testing.

Step 3: GitHub Actions CI/CD Pipeline Configuration

Cloud Automation

Configured `.github/workflows/ci-cd.yml` across two distinct stages: (1) **Test Stage:** spins up a Node.js matrix (20.x, 22.x) to execute syntax linting and tests; (2) **Build & Push Stage:** initializes QEMU and Docker Buildx, authenticates with Docker Hub via repository secrets (`DOCKERHUB_USERNAME`, `DOCKERHUB_TOKEN`), and pushes multi-architecture image layers with GitHub Actions cache (`type=gha`).

Step 4: Kubernetes Manifests & Local Minikube Deployment

Kubernetes Orchestration

Authored declarative manifests (`k8s/deployment.yaml`, `k8s/service.yaml`). Configured 2 pod replicas, rolling update strategy (`maxSurge: 1`, `maxUnavailable: 0`), resource quotas, and HTTP liveness/readiness probes targeting `/health`. Automated deployment using `scripts/deploy-minikube.sh` with Minikube service exposure on NodePort `30080`.

Step 5: End-to-End Verification & Real Screenshot Capture

Verification & Validation

Verified pipeline execution on GitHub: Workflow Run `#42349368109` achieved 100% green status across all jobs. Pushed image tags to Docker Hub repository `naveenkumar21032006/cicd-web-app`. Verified Minikube cluster pods active on macOS and captured high-resolution browser screenshots directly at `http://localhost:3000`.

GITHUB ACTIONS RUN	DOCKER HUB REGISTRY	LOCAL MINIKUBE CLUSTER
SUCCESS	**PUBLISHED**	**2/2 READY**
Run #42349368109 Passed	Tags: latest & sha-7c90fa7	Active at localhost:3000

5 Conclusion

This project proves that robust, enterprise-grade Continuous Integration and Continuous Deployment workflows can be established without cloud expenses. By leveraging GitHub Actions' cloud-hosted runners for testing and container compilation alongside free public registries (Docker Hub) and local container runtimes (Minikube / Docker), developers gain full visibility into automated delivery pipelines.

The pipeline achieved zero build errors, zero runtime vulnerabilities, sub-second test execution, and zero-downtime rolling updates verified through Kubernetes readiness checks. This architecture serves as an optimal blueprint for student projects, DevOps portfolio demonstrations, and local development environments seeking industry-standard practices without infrastructure billing.